



MSc. Thesis example M2 Health-AI

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Abstract

Cognitive radio (CR) is an emerging technology aimed at improving spectrum utilization efficiency. Spectrum sensing (SS), involving the detection of primary users (PUs) within a frequency band, stands as a critical cognitive radio task, facilitating bandwidth allocation to secondary users for communication purposes. While recent researches have explored spectrum sensing through artificial intelligence, assessed via computer simulations, the challenges posed by real-world environments, often characterized by low signal-to-noise ratios (SNR) and correlated noise, must be acknowledged. To address this, we propose an experimental approach involving an eight-receiver multireceiver system employing two software-defined radios (SDRs). This approach evaluates an artificial neural network model founded on covariance absolute value (CAV) and logarithmic determinant (logdet), bridging the gap between simulation-based studies and real-world complexities. Our findings indicate that combining multireceivers with a machine learning algorithm enhances detection probability.

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Glossary

SDR	Software Defined Radio
CR	Cognitive Radio
CAV	Covariance Absolute Value
PU	Primary User
ANN	Artificial Neural Network
FFNN	Feed Forward Neural Network
MIMO	Multiple Input Multiple Output
BBPN	Backpropagation Neural Network
SNR	Signal to Noise Ratio
SVM	Support Vector Machine
Logdet	Logarithmic Determinant
USRP	Universal Software Radio Peripheral
MFFNN	Multi-Layer Feed-Forward Neural Network
SU	Secondary User
QOS	Quality of Service
RF	Radio Frequency
ML	Machine Learning
AI	Artificial Intelligence
ROC	Receiver Operating Characteristic
SS	Spectrum Sensing
MF	Matched Filtering
FPGA	Field-programmable Gate Array
ED	Energy Detection
P_d	Probability of Detection
P_{fa}	Probability of False Alarm
MLP	Multilayer Perceptron

General Introduction

My Master 2 internship took place at Université Gustave Eiffel in the LEOST laboratory (Laboratoire Ondes et Signaux pour les Transports), from COSYS (COMPOSANTS ET SYSTÈMES) department. The work was performed in the context of the X2Rail-4 project. It is a European project within the Shift2rail program (https://projects.shift2rail.org/s2r_ip2n.aspx?p=X2RAIL-4).

The project is about the development of reconfigurable Wireless Sensor Networks (WSNs). The solution developed by Université Gustave Eiffel consists in the deployment of different sensors using various wireless technologies. A node onboard the train is able to collect and forward data, whatever the technology is. This node is equipped with cognitive radio features. The solution is called NEWNECTAR (New Generation of Adaptable Wireless Sensor Network for Wayside Objects in Railway Environments). This solution makes use of SDR (software Defined Radio) technology to create a Universal Sink Node (USG) embedded on the train that can sense the frequency spectrum and change the radio technology used to be connected, based on radio conditions and the presence or not of the technology in the environment. Each USG gets instructions from [7]. It also uses a technology called Software Defined Networking (SDN) to manage and control the wireless network more effectively [7].

The internship focused on spectrum sensing function of the cognitive radio device. The spectrum sensing aims at identifying unused portions of the frequency spectrum to allow unlicensed users (secondary user) to access. This helps ensure that licensed users (primary user) are not disturbed. In the context of the project, the aim is to enable the train as a secondary user to find a suitable frequency band for communication with the sensors in the neighborhood. The objectives are twofold: enhancing the existing SDR-based spectrum sensing test bed, and developing an Artificial Neural Network (ANN) model using the extracted features the Covariance Absolute Value (CAV) [8] and the Logarithmic determinant (Logdet) [9]. The ANN model will be designed to more accurately predict the presence of primary users, which in turn improves spectrum sensing, reduces interference, and enhances overall communication efficiency [10].

The structure of the report is outlined as follows. Chapter 1 provides an introduction presenting the background and significance of cognitive radio and spectrum sensing. Chapter 2 will delve into a discussion of the pertinent existing research conducted in this field. Moving on to Chapter 3, the methods and materials employed in this study will be elaborated upon. Chapter 4 will showcase the outcomes acquired from the research. Finally, the report will conclude by summarizing the findings and offering insights into the potential implications and future directions of this work.

Chapter 1

Cognitive Radio and Spectrum Sensing

1.1 Introduction

Modern rail transportation relies heavily on wireless communication, but the movements of trains degrade the wireless communications between the trains and the ground. In addition, the deployment of various wireless systems to answer train control and automation needs has conducted to frequency spectrum scarcity. The use of cognitive radio features in railway wireless systems is an active research field, with most studies focusing solely on solving the issue of spectrum scarcity [11].

Close to the railway, the access network directly serves data to trains, ensuring high-speed access via various wireless technologies for seamless services. The train network offers high-speed communication through a specialized terminal, connecting via external antennas. Signals reach gateways, enabling data access on train cars or the control terminal. Consequently, high-speed trains traverse through numerous regions encompassing distinct base station areas, each governed by varying device regulations. This situation gives rise to irregular and inconsistent frequency gaps experienced during the journey. To address these issues, a new cognitive architecture is crucial for reliable train communication amongst varying frequencies [11].

1.2 Cognitive Radio Concept

In the present telecommunication landscape, each system operates within a fixed frequency band assigned to it in the electromagnetic spectrum. While some emerging technologies have the potential to enhance service quality (QoS) and through-puts without necessitating increased bandwidth, certain frequency bands have become saturated over time. However, it is observed that the entire spectrum is not uniformly utilized across both time and space. Specific frequency bands may remain unexploited by users due to their fixed allocation in the spectrum, resulting in accessibility issues.

To tackle this pressing concern, the concept of Cognitive Radio (CR) was introduced by J. Mitola in 1999 [12]. Regarded as a technology with substantial potential in facilitating the implementation of opportunistic spectrum access, cognitive radio (CR) is recognized as an intelligent wireless communication system. This system demonstrates an awareness of its immediate environment, acquires knowledge from its surroundings, and dynamically adjusts its operational parameters in real time [13].

Therefore, we can illustrate the cognitive radio in three definitions:

- **Awareness:** the CR must be aware of itself and its operating environment. This may include that the CR is aware of the radio frequency (RF) environment, the rules that govern its operation, priorities and authorities, its user's needs, its geographic position, *etc.*
- **Adaptation:** this is the ability to adapt to the environment or user needs. For instance, the CR can adjust its behavior by changing its power, modulation, waveform strategy, carrier, and signal bandwidth. This level of adaptation recalls the agility of the SDR that enables flexible and adaptable radio systems by using software to replace conventional hardware components.
- **Cognition:** it is the capability of understanding and learning from its environment, its inner state, the impact of its actions, and past experiences [14].

While spectrum is typically licensed for extended use, a significant portion remains underutilized due to sporadic usage, wasting valuable frequency resources. To address this, the US FCC has allowed unlicensed device use in licensed bands. This led to the proposal of dynamic spectrum access techniques, aiming to enhance efficiency. This emerging research area focuses on improved cognitive radio (CR) technology with a soft-defined radio (SDR) platform to address railways' needs, where unlicensed devices (secondary users or SUs) share channels with licensed devices (primary users or PUs). CUs monitor for PUs and adjust settings accordingly to optimize spectrum usage [11].

1.3 Spectrum Sensing

The scarcity of available radio spectrum has emerged as a pressing concern in the field of future network research, requiring immediate attention [15].

Employing cognitive radio technology, a topic that has been thoroughly studied by researchers for almost two decades [16], could be a solution to these and similar problems. Wireless devices can now use cognitive radio technology to perceive the radio spectrum, evaluate the state of the frequency channels, and modify their communication settings to meet quality-of-service requirements while using less energy [12]. These devices have the ability to use both licensed and unlicensed frequency bands, but only when the main licensed band users are not using them. This prevents unwanted interference.

Wideband and narrowband spectrum sensing techniques are the two main categories of sensing methods that have been proposed in the past years. Narrowband sensing involves examining individual frequency channels, whereas wideband sensing simultaneously assesses multiple frequencies. Some of the most important narrow band sensing techniques are energy detection, cyclostationary feature detection, matched filter detection, covariance-based detection, and machine learning-based sensing [15]. We can find comparison of various sensing methods in [17].

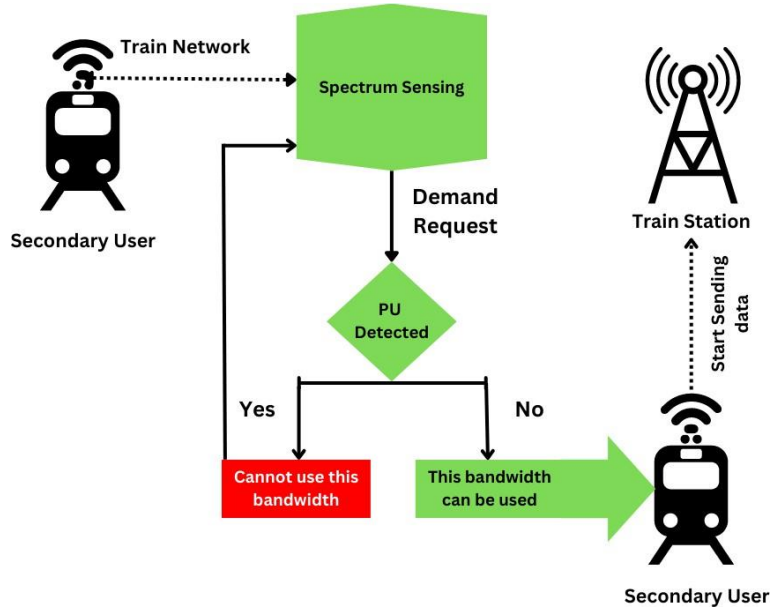


Figure 1.1: Spectrum sensing scenario

In the upcoming sections, I will present the well-known spectrum sensing techniques without machine learning, and in the second part including AI-based sensing and covariance-based detection, which will be subsequently employed in the project I am currently engaged in.

For the rest of this report, we denote by N the number of collected samples, M is the number of receiving (Rx) channels, L is the number of training blocks, $\log(\cdot)$ denotes the natural logarithm, $\det(\cdot)$ refers to the determinant, and $*$ is the convolution operator.

1.3.1 Spectrum Sensing System Model

In spectrum sensing, a decision should be taken regarding the availability of a specific frequency band. The decision is made to reject H_0 in favor to H_1 . H_1 represents the hypothesis that there is a signal from a primary source, while H_0 suggests that there is only background noise.

Consider the scenario where $s(t)$ represents the signal emitted from a sole antenna, $r_m(t)$ denotes the received signal, $h_m(t, \tau)$ signifies the impulse response of the channel (answer of the channel to a direct function), and $\eta_{m(t)}$ stands for the noise signal, each of these relating to the m th receiving antenna. The subsequent binary hypothesis has the potential to streamline the challenge of spectrum sensing:

$$r_m(t) = \begin{cases} \eta_m(t), & \text{under } H_0 \\ h_m(t, \tau) * s(t) + \eta_m(t), & \text{under } H_1 \end{cases} \quad (1)$$

The matrix that captures the covariance of the received signal is expressed as follow

$$R_{ss} = \begin{cases} \Theta, & \text{under } H_0, \\ R_{hs} + \Theta, & \text{under } H_1 \end{cases} \quad (2)$$

In the context of a situation involving a matrix $\mathbf{R}_r$ of size $M \times M$, where $\mathbf{R}_{hs}$ represents a noise-free sample covariance matrix derived from received signals, accounting for impairments caused by the radio channel. Θ denotes the covariance of noise present in received samples. It's worth noting that in practical scenarios involving multiple antenna receivers, the noise samples are interconnected. This means that Θ doesn't maintain a diagonal structure due to noise correlations. Despite this noise correlation, the matrix $\mathbf{R}_r$ still exhibits dissimilarity under the H_0 hypothesis compared to the H_1 hypothesis as indicated by equation (2). This distinction forms the basis for employing machine learning techniques to differentiate between the H_0 and H_1 hypotheses [18].

1.3.2 Spectrum Sensing Techniques without Machine Learning

Spectrum sensing core goal is to detect unused frequency bands, known as spectrum holes, while minimizing disruption to primary users. Various methods are used for this purpose, including energy detection, matched filter detection, and covariance absolute value. In the following subsection, we explore these spectrum sensing techniques, uncovering their principles and significance in the context of cognitive radio technology [10].

A. Energy Detection

The Energy Detection (ED) algorithm is a widely used and simple non-coherent detection method in cognitive radio systems. It has the advantage of not requiring prior knowledge of primary user signals and has low computational complexity [19]. It compares the energy of the received signal within a specific frequency band to a threshold, regardless of the signal structure, to make a detection decision. However, ED has limitations such as poor performance in low signal-to-noise ratio scenarios, difficulty in distinguishing PU signals from interference, and most important its sensitivity to noise uncertainty. Nonetheless, it remains popular and useful in cognitive radio due to its simplicity and applicability in various environments [20].

With a given number N of sampled signals $Y_i(n)$, the ED method calculates the total signal power denoted as E and compares it with the threshold λ_{ED} to judge the presence or absence of PU. The ED method performs the hypothesis testing according to

$$E = \sum_{i=1}^M \sum_{n=0}^{N-1} |Y_i(n)|^2 \quad (3)$$

$$\begin{cases} E, & \text{if } E > \lambda_{ED}, H_1 \\ 0, & \text{if } E \leq \lambda_{ED}, H_0 \end{cases} \quad (3.1)$$

H_0 is the hypothesis that primary users do not exist, and H_1 is the hypothesis that primary users exist. In the context where $|x|$ represents the norm operation of a complex number x , N is the number of sampled signals $Y_i(n)$ received by the i th antenna, M is the number of antennas, and λ_{ED} is the threshold of energy detection (ED) [21].

B. Matched filtering

A matched filter (MF) is a coherent detection technique used in communication theory to maximize the signal-to-noise ratio in channels affected by additive white Gaussian noise (AWGN) [22]. It is considered an optimal detector in such scenarios. Coherent detection requires the secondary user (SU) to have prior knowledge of both the physical parameters of the primary user's system and the structure of the primary signal [23]. By cross-correlating a known sequence with the received signal, the MF can efficiently detect the presence of a primary signal when a significant correlation peak is observed. Conversely, the absence of a correlation peak indicates a vacant band. This approach leverages the coherence between the received signal and the known sequence for effective primary signal detection [24].

With a given number N of sampled signals $Y_i(n)$, the matching filter method calculates the correlation between the received signal $Y_i(n)$ and the reference signal $h(n)$ for each antenna i , resulting in the matched filter output E_i for the i th antenna. This output is compared with a threshold λ_{MF} to determine the presence or absence of PU. The matching filter method performs the hypothesis testing according to

$$E_i = \sum_{n=0}^{N-1} Y_i(n) \cdot h(n)^* \quad (4)$$

$$\begin{cases} E_i, & \text{if } E_i > \lambda_{MF}, H_1 \\ 0, & \text{if } E_i \leq \lambda_{MF}, H_0 \end{cases} \quad (4.1)$$

Where H_0 is the hypothesis that primary users do not exist, and H_1 is the hypothesis that primary users exist. In this context, $|x|$ represents the norm operation of a complex number x , N is the number of sampled signals $Y_i(n)$ received by the i th antenna, M is the number of antennas, $h(n)$ is the reference signal, and λ_{MF} is the threshold of the matching filter method.

C. Covariance Absolute Value

Covariance Absolute Value (CAV) detection is a robust non-parametric method. Its effectiveness stems from its ability to leverage the uncorrelated nature of noise observed among sensors, achieved through the examination of off-diagonal elements in the sample covariance matrix. Thereby maintaining independence from Signal-to-Noise Ratio (SNR) variations, ensuring its resilience in handling noise uncertainty over energy detection. Making it a preferred choice for spectrum sensing and signal processing tasks [25].

The CAV algorithm, introduced by Zheng and Liang [26], effectively operates by analyzing the ratio of the absolute sum of entries in the sample covariance matrix to the sum of its diagonal entries. This ratio tends to approach 1 when there is no correlation between signals (null hypothesis) and exceeds 1 when there is a correlation as stated in equation (2).

The equation of the covariance matrix is:

$$Cov = \frac{\sum_{i,j}^M |R_r(i, j)|}{\sum_i^M |R_r(i, i)|} \quad (5)$$

In this context, M signifies the count of variables or dimensions within the dataset. The non-diagonal elements, denoted as $R_r(i, j)$, depict the correlation coefficients between variables i and j and the diagonal elements, represented as $R_r(i, i)$.

1.3.3 Spectrum Sensing Techniques with Machine Learning

As highlighted earlier, several conventional methods are employed for spectrum sensing (SS), each necessitating a distinct threshold to effectively detect primary users (PUs). Determining these thresholds involves the application of distinct algorithms for each method. Setting the threshold too high can lead to missed detection, while setting it too low can result in false alarms.

To address the challenges associated with these conventional SS methods, cognitive spectrum sensing (CSS) is employing machine learning (ML) frameworks. ML algorithms can handle complex problems by analyzing data patterns and structures, as they possess the ability to learn, reason, and make decisions. The integration of ML algorithms into cognitive radio networks enhances their ability to adapt, automate processes, and increase network flexibility. In recent times, three types of ML algorithms—unsupervised, supervised, and reinforcement learning (RL) [27] have been used for spectrum sensing. Although various traditional SS techniques remain in use, the introduction of ML aims to enhance detection accuracy and overall system performance [1].

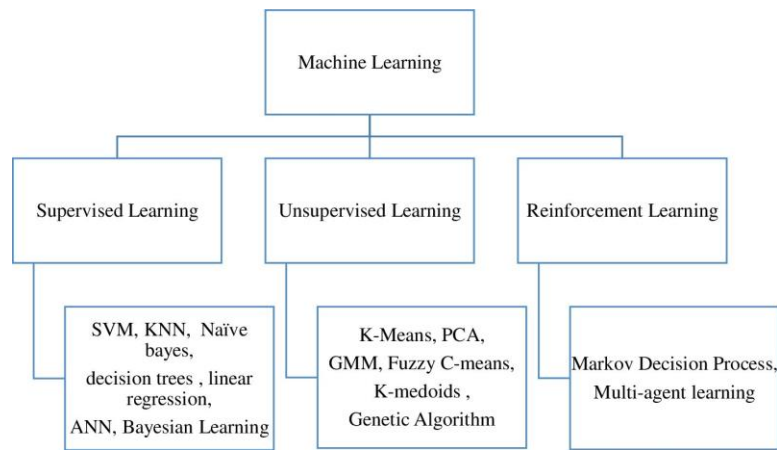


Figure 1.2: Machine Learning Algorithms for SS [1]

Supervised and Unsupervised Learning

Unsupervised learning is a type of machine learning in which the algorithm is trained using unlabeled data. Unlabeled data means that the input data is not pre-assigned with any output labels or categories. The algorithm must identify underlying patterns or relationships in the input data without any external

guidance or supervision. Clustering, dimension reduction, and anomaly detection are some examples of unsupervised learning techniques [28].

On the other hand, supervised learning is a type of machine learning in which the algorithm is trained using labeled data. Labeled data means that the input data has been pre-assigned with corresponding output labels or categories that the algorithm should learn to predict. The algorithm is trained on this labeled data to learn the underlying patterns or relationships between the input data and the output labels. Once trained, the algorithm can then use the learned patterns to predict the output labels for new, unseen input data [28].

Unlike the unsupervised learning techniques that may be used in unknown environments without having any prior knowledge, supervised learning techniques are generally used in certain familiar/known environments, with prior knowledge about the characteristics of the environment. In the following, we introduce some of the major supervised learning techniques that have been applied to the cognitive radio literature [29].

Chapter 2

Background

2.1 General Introduction

The electromagnetic spectrum, a valuable natural resource, has been allocated to different wireless services by the governmental authorities. However, many frequency bands in the spectrum remain unoccupied or partially used, despite the increasing number of wireless devices and applications worldwide. Cognitive radio (CR) is a proposed solution to the under-utilization problem. Spectrum sensing, the key component of CR, aims to detect unused frequency bands (spectrum holes) without interfering with primary users. Various detection methods, such as energy detection, matched filter detection, and covariance absolute value have been discussed [10].

2.2 Artificial Neural Network

The exploration of Artificial Neural Networks (ANNs) has been driven by the recognition that the human brain processes information fundamentally differently from conventional digital computers, as highlighted in Haykin's work [30]. An ANN is essentially a highly parallel distributed processor composed of simple processing units, with a natural capacity for acquiring experiential knowledge and making it accessible for use [30]. ANNs share two key similarities with the human brain: firstly, they acquire knowledge from their environment through a learning process, and secondly, they use interneuron connection strengths, referred to as synaptic weights, to store this acquired knowledge.

Some of the notable advantages and capabilities of ANNs include their ability to: 1) model nonlinear relationships that align with underlying physical processes, 2) adapt to minor environmental changes, and 3) in the context of pattern classification, provide not only information about which pattern to select but also the confidence associated with that decision. [29]



Figure 2.1: ANN Classification

The manifold advantages of this technology have led to its extensive implementation across various domains, enhancing decision-making and predictive capabilities. Furthermore, it provides plenty of methods and techniques that can be adeptly utilized to tackle a diverse array of challenges. Among these techniques, the multi-layer feed-forward neural network stands out as a particularly vital and influential tool for surpassing the limitations of conventional spectrum sensing detection methods. [31]

The performance of the SS implementation can be characterized using two probabilities: P_d the probability of detection and the P_{fa} false alarm probability. Balancing these probabilities is crucial in designing robust detection systems [6].

P_d is the probability of detecting a signal in the band of interest when this signal is truly present. Failed detection causes interference with the PU.

$$P_d = P(\text{decision}H_1/H_1) \quad (5)$$

P_{fa} is the probability that the test incorrectly decides that the band of interest is occupied when it is free. The False alarm reduces the efficiency of spectrum use.

$$P_{fa} = P(\text{decision}H_1/H_0) \quad (6)$$

Within cognitive radio (CR) systems, the primary necessity for spectrum sensing revolves around promptly and precisely identifying the presence of the primary user. Following extensive analyses, it became evident that employing an artificial neural network (ANN) for spectrum sensing is suitable for detecting signals even in significantly low signal-to-noise ratio (SNR) conditions, exhibiting dependable performance. Simulations further demonstrate the superiority of the proposed technique compared to prior methods when conducting spectrum sensing for CR systems within noisy environments [32].

The NBC algorithm uses Bayes' theorem with independence assumptions between the feature inputs to procure decisions. Each feature input is given the same importance, which means that each attribute contributes equally to the decision. The main advantage of the NBC algorithm is the least computational time required for training the dataset using high dimensional features with limited data [26].

[2] Introduces new machine learning-based blind spectrum sensing methods to improve PU detection in cognitive radio networks, especially in vehicle-to-everything communication V2X which face severe multipath environments. In which multipath refers to a phenomenon in wireless communication where transmitted signals take multiple paths to reach the receiver due to reflections, scattering, and diffraction. These paths can lead to the reception of multiple versions of the same signal at slightly different times

and amplitudes. Multipath propagation can result in signal fading, distortion, and interference, making accurate spectrum sensing challenging, as it introduces fluctuations in signal strength and characteristics. Proper handling of multipath effects is crucial in spectrum sensing algorithms to ensure reliable detection and interpretation of primary user signals in the presence of such signal variations caused by multipath propagation. The proposed ML approaches (Naive Bayes Classifier, Support Vector Machine [18], and Artificial Neural Network) outperform traditional SS methods. These ML-based detectors show promise for PU detection over multipath channels, particularly in low SNR indoor scenarios since the major weakness of the classical fixed threshold is the rate of wrong detection in multipath channels.

Based on a comparative analysis of ML strategies and eigenvalues, this study [2] concludes that ML models, particularly the ANN approach, outperform other methods in spectrum sensing including blindly combined energy detection (BCED) [33], maximum-eigenvalue-geometric-mean (MEGM) [34], maximum-minimum eigenvalue (MME) [35] as presented in the following figure.

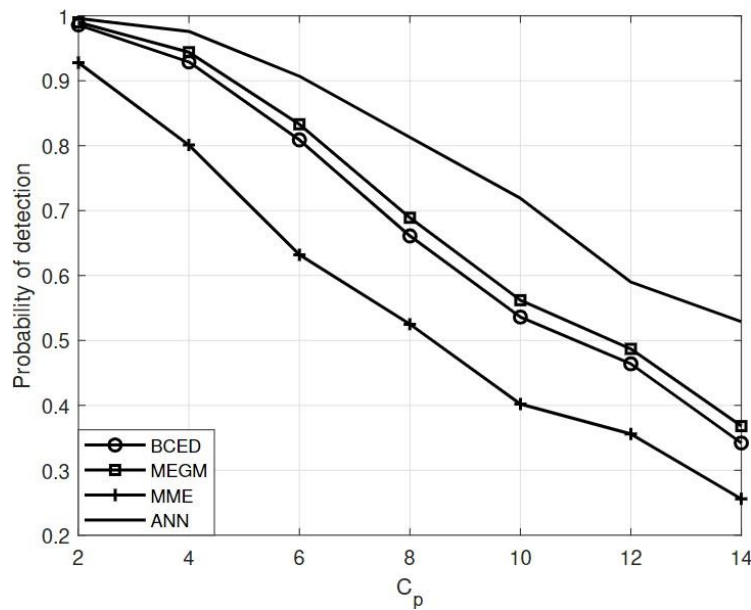


Figure 2.2: Probability of detection versus multipath [2]

Significantly, the ANN method exhibits the highest level of performance, showcasing resilience against the multipath effect, unlike eigenvalues which are sensitive and experience performance degradation with increasing channel order. These findings highlight the effectiveness of ML-based sensing algorithms and emphasize the potential of ANN models in addressing the challenges of modern wireless communication systems.

Overall [2] shows that ML approaches, especially the ANN method, offer improved spectrum sensing capabilities and show promise for practical implementation in real-world scenarios. Present the (clinical) context and the state-of-the-art.

2.2.1 Supervised and Unsupervised Learning Techniques

In [29] the authors explore channel estimation in Cognitive Radios (CRs) and Cognitive Radio Networks (CRNs), focusing on highly reconfigurable Software-Defined Radios (SDRs). The complexity arises from the simultaneous adjustment of multiple parameters such as transmit power, coding scheme, modulation scheme, and sensing algorithm that interact and influence the RF environment. Learning methods like threshold learning algorithms allow CRs to adapt without full knowledge of parameter inter-dependencies.

To achieve configurability and autonomous operation in unknown RF environments, CRs use learning algorithms to adapt and coordinate with other radios. To reduce system complexity, learning algorithms with low complexity are preferred. Recent research investigates supervised and unsupervised learning techniques, including neural networks and support vector machines (SVMs), for various CR learning tasks.

This paper highlights the importance of autonomous learning in CRs and presents a survey of machine learning techniques applied to CRs. Practical applications of these techniques are demonstrated through two examples, one using Artificial Neural Networks (ANNs) and the other using Case-Based Reasoning (CBR), to demonstrate practical applications of these techniques. The paper discusses supervised and unsupervised learning paradigms and evaluates the advantages and limitations of different machine-learning methods for CRs in different contexts and tasks.

The model described in [3] employs a multilayer perceptron (MLP) neural network to illustrate the operation of ANN in CR contexts. Simulations reveal that the number of hidden layers affects convergence speed and mean squared error (MSE), requiring a trade-off between speed and accuracy based on available training data.

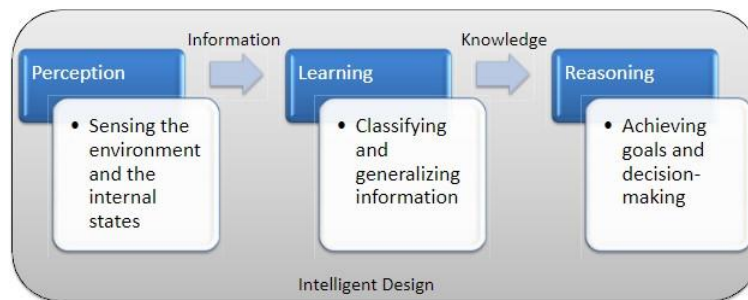


Figure 2.3: Transforming Information into Knowledge: The Power of Intelligent Learning [3]

2.2.2 Feed-Forward Neural Network

Multi-layer Feed-Forward Neural Network

Cognitive radio (CR) faces two main challenges: making effective decisions and accurately classifying features. To tackle these hurdles, we've presented a model based on a Multi-layer Feed-Forward Neural Network. This model aims to enhance decision accuracy across varying noisy conditions and has been implemented through a Neural Network system (NNS), significantly boosting the desirable throughput. Neural Networks exemplify intelligent behavior, equipped to comprehend the surroundings, learn, and

adapt real-time operational parameters based on the distinct requirements of unlicensed users. In the following sections, we will delve into the details of how this model enables us to surmount these challenges.

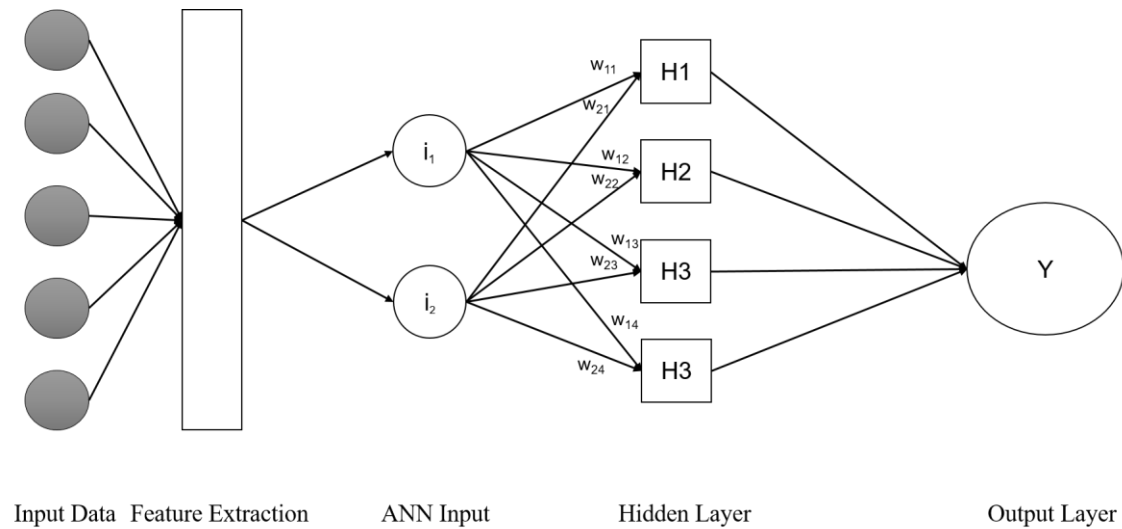


Figure 2.4: Multi-layer ANN

Figure 2.4 illustrates the overall structure of the Multi-layer Neural Network with first extracting the important features.

The input layer receives information intended to elucidate the analysis problem and distributes it to the neurons within the input layer.

The hidden Layer, is positioned between the input and output layers, this intermediary layer enables neural networks to replicate nonlinear processes. The outputs from each hidden layer serve as inputs for the units in the subsequent layer.

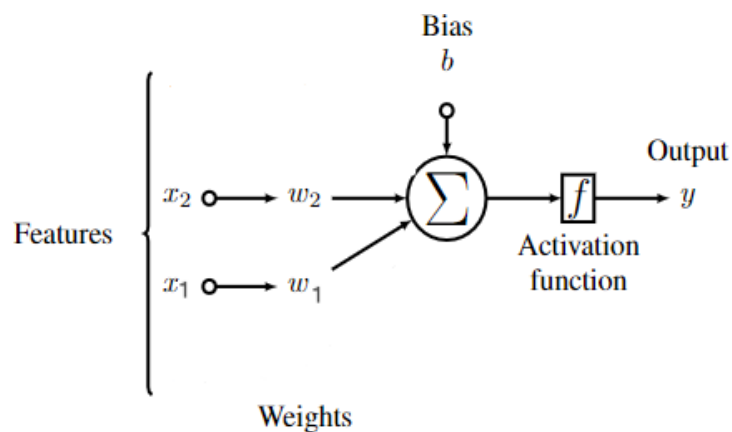


Figure 2.5: Hidden layer neuron structure

The output layer is the last layer of the network, it is responsible for generating predictions and results.

Below is a concise explanation of MFNNs. To delve deeper into the topic, readers are encouraged to explore the extensive body of literature on Neural Networks (for instance, refer to [36]- [37]). Feed Forward Back Propagation entails two distinct passes: the forward pass and the backward pass. [38] Figure 3.7 shows our ANN model that was built and that will be discussed in the following section.

At the core of the Forward pass lies the individual neuron or perceptron, which establishes the following connection between its inputs $\mathbf{x}_i$, where i ranges from 1 to M (the number of antennas), and its resultant output y :

$$y = f(a), \quad a = \sum_{i=1}^M w_i x_i + b \quad (7)$$

Here, w_i represents the weights corresponding to each input, b stands for the bias, and $f(a)$ denotes the activation function. In our scenarios, the activation function is a sigmoid function. To make the neural network bias free (neutral for the binary hypothesis problem), the training dataset consists of an equal amount of primary signal feature examples and noise feature examples.

The sigmoid function is a widely used activation function in neural networks due to its differentiability and non-linearity. It has a smooth gradient that leads to faster convergence during training and is suitable for optimization algorithms. Additionally, the sigmoid function has a probabilistic interpretation and is often used in binary classification tasks. Despite some drawbacks, such as saturation in neurons and computational expense, the sigmoid function remains an important tool in the neural network toolbox for its versatility and optimization capabilities. It is an “S” curve that generates an output between 0 and 1, it is expressed as a probability. The sigmoid function’s formula is:

$$f(a) = \frac{1}{1 + e^{-z}} \quad (8)$$

In [4], the authors presented a novel approach that combines spectrum sensing technology with Artificial Neural Networks (ANN) to enhance the efficiency and accuracy of spectrum sensing in cognitive radio networks.

This paper discusses the limitations of energy detection and the need for prior knowledge of the primary user signal in parametric features. So a unique and innovative hybrid spectrum sensing scheme has been introduced. The effectiveness of this approach has been assessed by conducting experiments using different radio technologies in a controlled test environment. This approach consists of using energy detection and the energy value from energy detection, along with Zhang statistic [39] from the likelihood ratio statistic [40] scheme, as sensing features for the ANN which is approached as a binary classification task. This approach aims to enhance the robustness of the system.

During the training phase, the data was labeled using specific features. It was trained using equal numbers of primary signals and AWGN feature examples. Four ANN architectures were tested across different radio technologies using 200,018 examples per technology. Training time for the ANN architecture was around 10-12 minutes.

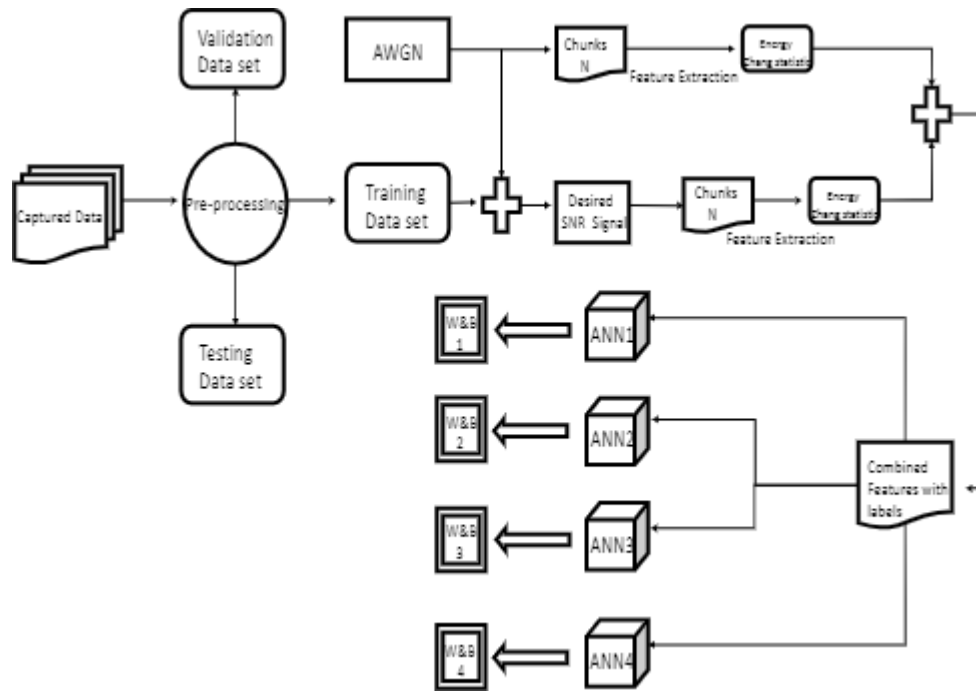


Figure 2.6: ANN Training [4]

After evaluating multiple procedures, the highest accuracy was obtained when they utilized the following four features, including two from the energy feature and two from the Zhang statistic. This combination of features yielded the best performance among all the tested configurations [4].

Table 2.1: Comparison of Results [4]

Radio Technology	Only Energy	Only Zhang Statistic	Energy and Zhang Statistic	All Four Features
FM Broadcast	76.66%	83.82%	84.15%	86.82%
GSM-900 DL	76.91%	83.93%	84.27%	86.88%
DCS-1800 DL	78.19%	86.49%	86.51%	89.04%
UHF TV	78%	85.97%	86.02%	88.55%

In [5] the author conducted an experimental evaluation of various strategies aimed at detecting the primary user and assessing the availability of frequency bands for secondary users. The obtained results are summarized in the following table.

Table 2.2: Average time delay in all techniques [5]

Cognitive User	Average Delay in All Techniques		
	Delay of FFNN	Delay of Underlay	Delay of Interweave
1	0.1	9	0.2
2	0.15	13.2	2.2
3	0.33	23.2	2.5
4	0.31	33.4	2.6
5	0.39	41.9	2.8
6	0.56	48.2	2.14
7	0.63	56.1	2.19
8	0.71	64.6	2.27
9	1.61	74.7	3.27
10	2.51	84.9	3.28

The findings presented in the table above demonstrate a comparison of the average time delay taken by feed-forward neural network (FFNN), underlay, and interweave. Where the time delay is the measure of the time required by the secondary user to get the position in the white band. The study revealed that the feed-forward neural network exhibited better performance than the underlay and interweave spectrum-sharing techniques in terms of reducing time delay. By training the neural network, it becomes capable of predicting the availability of primary users in the future, allowing efficient placement of secondary users within the vacant frequency bands. The advantages of employing neural networks for spectrum sharing in cognitive radio can be summarized as follows:

1. **Faster Band Availability Prediction:** Neural networks require less time to predict the availability of frequency bands compared to underlay and interweave techniques.
2. **Higher Throughput:** Feed Forward Neural Network spectrum sharing achieves greater throughput compared to other sharing techniques.
3. **Cost-Effectiveness:** Implementing a Feed Forward Neural Network for spectrum sharing incurs lower real-world adaptation costs. The required machine learning tools can be fully established through software, eliminating the need for additional hardware, and thus reducing expenses.
4. **Periodic Data Updates:** Spectrum sharing using a feed-forward neural network necessitates periodic training to ensure the model stays up to date with any changes in primary user behaviors. This allows the system to adapt to dynamic spectrum conditions.

The author in [6] discusses the widespread use of the energy detection (ED) method for spectrum sensing, which offers simplicity and does not require prior knowledge of the primary user (PU) signal. Additionally, the author explores the application of artificial neural networks (ANN) as a soft computing technique for PU signal detection. This paper introduces a system that utilizes a Raspberry Pi 3, a 433 MHz Wireless transmitter, an RTL-SDR receiver, and MATLAB software to transmit a real signal, which can then be detected by an RTL-SDR dongle connected to MATLAB-SIMULINK using the energy detection method and artificial neural networks (ANN).

The author employed a Feedforward Network as an artificial neural network (ANN) model and utilized the Backpropagation algorithm to adjust the weights and minimize the error. The table provided displays the probability of detection and the false alarm probability values acquired through the energy detection (ED) detector.

Table 2.3: The probability of detection P_d and the false alarm P_{fa} probability of ED [6]

P_d	P_{fa}
0.98 %	0.015 %

In paper [6], the author undertook a study where they compared the performance of two training methods: 'trainlm' (Levenberg Marquardt) and 'trainscg' (Scaled Conjugate Gradient). The outcomes, displayed in a tabular format, distinctly demonstrate that the 'trainlm' function with 8 hidden neurons yielded the most advantageous results.

The Levenberg-Marquardt backpropagation (trainlm) algorithm is employed to minimize a multivariate function expressed as the sum of squared values of nonlinear functions. It operates iteratively, ensuring a reduction in the performance function at each step, which enhances its efficiency for moderately sized networks. The algorithm's key equation involves a combination of the gradient and an approximated Hessian matrix. However, similar to trainbfg, this advantage comes with the trade-off of increased memory usage and computational overhead [41]. While the Scaled Conjugate Gradient (trainscg) algorithm differs from other conjugate methods by not requiring a line search in each iteration. It employs a step size scaling mechanism that speeds up the process compared to other second-order algorithms. While it might take more iterations to converge, the reduction in computation per iteration is substantial as no line search is performed [41].

Table 2.4: Probability of detection P_d and the false alarm probability P_{fa} in testing phase [6]

Number of neuron in the hidden layer	Training function	P_d	P_{fa}
3	SCG	0.983	0.017
	LM	0.963	0.020
4	SCG	0.973	0.013
	LM	0.966	0.043
5	SCG	0.980	0.0167
	LM	0.960	0.046
6	SCG	0.970	0.0167
	LM	0.973	0.0167
7	SCG	0.963	0.0267
	LM	0.960	0.023
8	SCG	0.970	0.0131
	LM	0.983	0.007
10	SCG	0.970	0.0233
	LM	0.956	0.0267

While the energy detection technique demonstrates strong performance in optimal situations, it encounters difficulties in scenarios with low signal-to-noise ratios due to uncertainties in noise, shadowing, and fading effects. The process of threshold estimation introduces intricacies to this approach. On the other hand, the Artificial Neural Network (ANN) detector surpasses the conventional energy detection method by providing improved and consistent outcomes.

Table 2.5: The probability of detection and the false alarm probability of Ann detector [6]

P_d	P_{fa}
0.983 %	0.007 %

Several spectrum sensing methods underwent testing, among which the energy detection approach and ANN exhibited the most favorable outcomes. Different combinations of hidden layer neurons were investigated in the case of ANN. The assessment of the suggested application focused on two key metrics: the probability of detection and the false alarm probability. The results indicated the ANN's dominance in terms of architecture, delivering a consistent ability for spectrum sensing when compared to the conventional energy detection technique.

2.2.3 Covariance Absolute Value in Neural Network

In [42], the writer constructed a neural network by utilizing the covariance matrix of the received signal as a feature for input in the model. The suggestion made in a prior work [43] involved employing the sample covariance matrix derived from the sensing samples as the input for the neural network. This approach was chosen due to the significant inclusion of vital energy-correlation details from the sensing samples within the sample covariance matrix. The results of this suggested technique showcased an impressive performance, achieving a detection probability of 96.7% at a false alarm probability of 1.9% when the signal-to-noise ratio (SNR) was set at -18 dB. This performance notably surpassed the capabilities of traditional methods.

2.2.4 Back-Propagation

During the training process, the ANN initially predicts outputs by processing the training dataset. This prediction is then refined using the back-propagation mechanism, which involves adjusting the network's weights to minimize prediction errors.

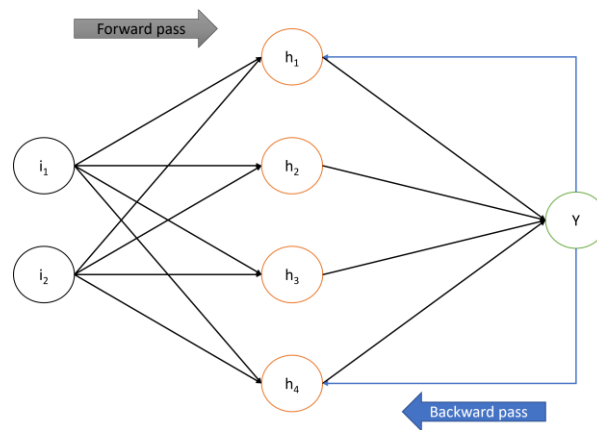


Figure 2.7: Forward pass and Backward pass

The significance of Backpropagation in feedforward binary classification lies in the iterative approach to improve predictions over successive iterations, often referred to as “epochs”. This is achieved by evaluating and minimizing error loss. By adjusting the network’s weights iteratively based on the discrepancies between predicted and actual outputs, Backpropagation empowers the network to progressively enhance its precision over time. Without Backpropagation, the network’s weights would remain static, lacking the adaptability needed to handle variations in input data [44].

Moreover, Backpropagation plays an important role in training neural networks equipped with nonlinear activation functions, such as the sigmoid function. Its strategic inclusion forms a fundamental aspect of advancing precise machine learning models. This strategy is complemented by the use of Binary Cross-Entropy as a loss function, particularly valuable for binary classification tasks. This approach is particularly effective when dealing with multiple classification problems that can be simplified into

binary decisions [45]. In our research, we emphasize the critical importance of selecting an appropriate loss function and the significant gradient descent method for training within the ANN model.

Gradient Descent, a widely utilized optimization approach in machine learning, adjusts parameters in the direction opposite to gradients to minimize the cost function. The learning rate holds a crucial role, dictating the step size and direction toward the minimum value.

When dealing with predictive models, the issues of overfitting and underfitting can have significant impacts on performance. Gradient Descent is applicable to both linear and nonlinear models. However, it's worth noting that it can be sensitive to local minima and saddle points. To enhance its efficiency, variations such as stochastic gradient descent are commonly used [46].

Backpropagation Algorithm [44]:

1. **Initialize Weights:** Initialize the weights of the neural network with appropriate values, usually randomly.

2. **Forward Propagation:** Enter the input from the training data (input, correct output) into the neural network. Compute the activations of each layer's neurons through the network's forward pass. Each neuron's activation is calculated by applying an activation function to the weighted sum of its inputs.

3. **Calculate Output Error:** Calculate the error (e) between the network's output and the correct output:

$$e = \text{correct output} - \text{network's output} \quad (10)$$

4. **Calculate Output Layer Delta (δ):** Compute the delta (δ) for the output layer using the derivative of the activation function (f') applied to the total input (v) of each output neuron:

$$\delta = f'(v) \cdot e \quad (11)$$

This delta quantifies how much each output neuron's input contributed to the overall error.

5. **Backpropagate the Error:** Propagate the error (δ) backward through the network layers. For each hidden layer: - Compute the delta (δ) using the derivative of the activation function and the weighted sum of deltas from the next layer. - The delta (δ) at a hidden layer represents how much the weighted sum of inputs to a neuron contributed to the error.

6. **Update Weights:** Using the calculated deltas (δ) at each layer, update the weights to minimize the error. For each weight (w_{ij}) connecting neuron j in the previous layer to neuron i in the current layer:

$$\Delta w_{ij} = \alpha \cdot \delta_i \cdot x_j \quad (12)$$

Where α is the learning rate, δ_i is the delta at neuron i , and x_j is the output of neuron j in the previous layer.

$$w_{ij} \leftarrow w_{ij} + \Delta w_{ij} \quad (13)$$

7. Repeat for Training Data: Iterate through the training data and repeat steps 2-6 for each data point.

8. Repeat Training Epochs: Repeat the entire process for multiple epochs (iterations) over the training data to improve the network's performance.

The paper [10] proposes a combined approach of spectrum sensing method using energy detection, cyclostationary feature detection, and artificial neural networks (ANNs) for detecting the primary user in a communication system. The chosen ANN model is the Backpropagation Neural Network (BPNN) with a hidden layer and an output layer. By training the BPNN using the "trainlm" training function, the weights and thresholds are adjusted iteratively to optimize the network's performance.

Four features are extracted from the signals: signal energy and three cyclostationary feature values. These features serve as inputs to the ANN, enabling it to learn and recognize patterns associated with the primary user's presence. The simulation results show that the detection performance is signal over noise ratio (SNR)-dependent, with high detection probability and low false alarm probability in low signal over noise ratio (SNR) environments. Achieving a detection probability of 100% is possible with a signal over noise ratio (SNR) of -10 dB or higher. The proposed method demonstrates stable performance with a consistently low false alarm probability of zero, utilizing a smaller sample size of 1024.

By setting the threshold of the cyclostationary detection method at 0.8 and adjusting it based on the training signal over noise ratio (SNR), the approach further improves the detection probability across different signal over noise ratio (SNR) levels. Overall, the paper presents a joint detection method using ANNs, offering superior performance in low signal over noise ratio (SNR) scenarios and reducing computational complexity while enhancing anti-interference capabilities. Future work involves expanding the application scope to simulate digital signals and incorporating pattern recognition techniques for identifying primary signal types and systems in cognitive radio systems.

Chapter 3

Material and methods

3.1 Experimental Platform

3.1.1 Transmitter Side

The study uses a primary user PU user signal that generates a 20-MHz LTE signal, and this signal is created using LabVIEW software. The hardware responsible for transmission includes a microcontroller linked to an NI-USRP 2944 board via an MXI-Express cable. This board incorporates a Kintex-7 field-programmable gate array (FPGA) and can accommodate a MIMO antenna configuration of 2×2 . The FPGA generates the baseband signal using a quadrature phase shift keying (QPSK) modulation approach. After its creation, the baseband signal is converted to a passband signal, amplified, and then sent over the air through the Tx/Rx port, as shown in the Figure below. For both the transmitter and receiver, Vert2450 Ettus dual-band omnidirectional RF antennas with vertical polarization are used in the experiment [18].

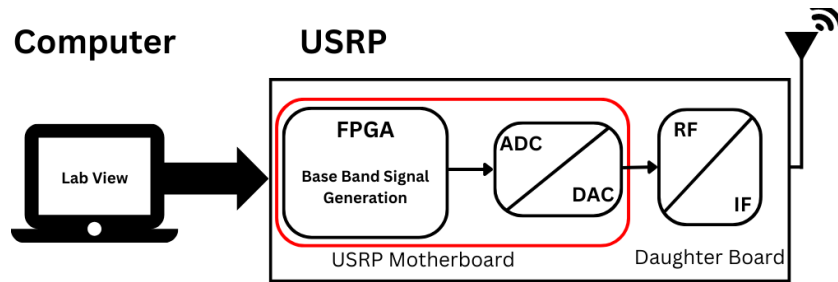


Figure 3.1: Transmitter Hardware

3.1.2 Receiver Side

On the recipient end, a host PC operating MATLAB is linked to the 1 Gigabit SFP+ port of two USRPs N310 through RJ45 cables. The primary role of the USRPs is to amplify, digitize, and downconvert the incoming signal. The N310 unit is outfitted with dual Ad9371 transceivers and can accommodate a MIMO antenna setup of 4×4 . [18] Therefore, two software-defined radios (SDRs) are employed, leveraging 8 receiver antennas, a measure taken to enhance the signal-to-noise ratio (SNR). To maintain uniform spacing of 2.5 cm between adjacent antennas, these antennas are arranged in a linear array on a dedicated holder. They are then connected to the USRP Rx ports via coaxial cables of equal length. The schematic diagram of the proposed platform is depicted in the upper part of the figure, while the actual utilized SDR devices are shown in Figure 2. The concept for this platform is influenced by the framework introduced in [47], which was employed to experimentally evaluate a machine learning-based algorithm for recognizing digital modulation.

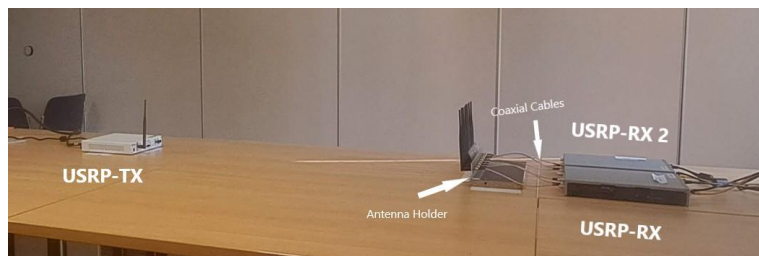


Figure 3.2: Proposed SDRs Platform

Receiver Side: Synchronization

For the simultaneous activation of the two receivers and the transmitter, achieving synchronization in terms of time, frequency, and phase is important. To achieve synchronization in time and frequency, an OCTOCLOCK module is utilized in conjunction with the USRP. This module generates a pulse-per-second (PPS) signal and a 10 MHz reference frequency that are employed by the receivers and the transmitter. In order to synchronize with the GPS clock, the OCTOCLOCK module's clock distributor is linked to a GPS antenna, which is positioned on the rooftop of a building to have an unobstructed view of the sky [18].

At the start of the experiment, we measure the initial difference in the way the transceivers handle phases. This is carried out by interconnecting all eight receiver channels to a signal source through a combination of two four-way RF splitters and a two-way RF splitter, which are connected using coaxial cables. Afterwards, we correct any differences in phase we found in the signals we receive as we continue with the measurements.

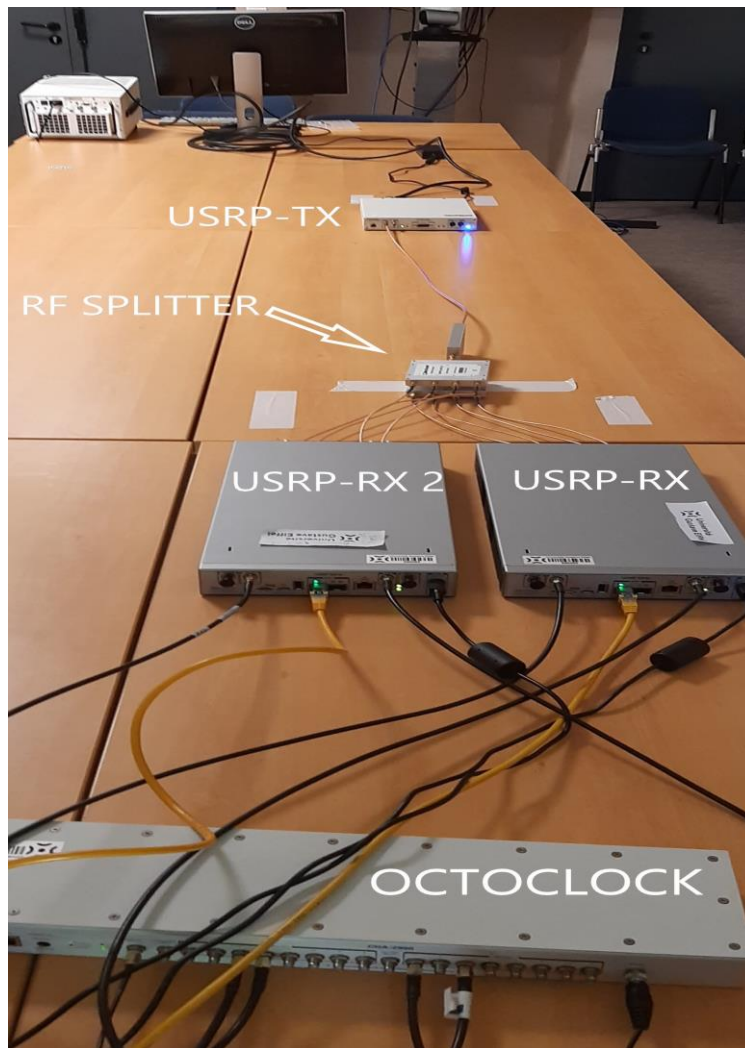


Figure 3.3: SDRs synchronization

Noise Signal Existence

To demonstrate the existence of noise correlation, we collected noise signal samples in our RF laboratory using the receiver of the platform depicted in Fig. 3.1, at the frequency of 5.23 GHz. During the acquisition process, we turned off the LTE signal generator, which is shown in Fig 3.2. The noise samples received by each pair of antennas over 1000 repetitions.

The successful implementation of synchronized SDRs and antennas, along with careful phase alignment, provided substantial improvements in the source-to-noise ratio and overall performance of the communication system. This enhanced setup holds promising potential for various applications in wireless communication and beyond. Further research and experimentation could explore additional techniques for even greater performance gains.

3.2 Machine learning Based solution

3.2.1 Artificial Neural Network (ANN)

The Multi-layer feed-forward neural network in our case is used as a binary classification.

Binary Classification

Within the scope of machine learning, a fundamental undertaking of paramount significance involves bi-nary classification. Given that our research is centered around the discernment of primary user occupancy within a bandwidth, this endeavor inherently aligns with the essence of binary classification.

There are two approaches to binary learning. The first approach involves using both positive and negative examples of a concept to learn how to tell them apart. The second approach focuses on using only positive examples of the concept to learn how to identify those specific instances. So binary classification can be achieved by supervised learning and unsupervised. But supervised methods are usually favored in the fields of Pattern Recognition, Machine Learning, Neural Networks, and Data Mining (e.g., Binary Hypothesis Testing (Fukunaga, 1990), Multi-Layer Perceptrons (MLP) (Rumelhart, Hinton, & Williams, 1986) [48]

For the input layer, we need to extract the features of the data. As we will discuss now.

Feature Extraction

Signal Acquisition: The initiation of the training phase involves obtaining $L/2$ noise signal blocks, each possessing dimensions of $M \times N$. These acquired data are stored within a three-dimensional array. This process occurs while the LTE generator remains inactive. Subsequently, the LTE signal generator is activated, marking the commencement of signal acquisition in the H_1 scenario. This continues until an additional $L/2$ PU signal blocks are obtained and stored in the same array alongside the noise data. This sequence culminates in the accumulation of L signal blocks, all set for progression to the subsequent module.

Feature Extraction: Following the acquisition of baseband signals, the construction of the feature vector transpires, utilizing the CAV [4] and logDet [11] test parameters in the subsequent manner:

$$x_{l=1}^L = \left\{ \frac{\sum_{i,j}^M |R_r(i,j)|}{\sum_i^M |R_r(i,i)|}, \log(\det(\mathbf{R}_r)) \right\}_{l=1}^L \quad (14)$$

The first and second $L/2$ blocks are assigned to the noise and PU signal classes, respectively, in the algorithm, referred to as LogDet-CAV-ANN.

Next, we move on to the ANN Classifier: Following the formation of the feature vector, we assign a label of 0 to the noise data and 1 to the PU signal data. Once the labeling is complete, the training set is prepared to be used as input for developing the ANN classifier. The primary goal of the ANN classifier is to create a model that can accurately classify data points into two classes.

The model has three phases: ANN training and ANN testing. The first two phases are the pre-phases and are conducted before the actual sensing starts. The scheme uses three data clusters training dataset, cross-validation dataset and testing dataset.

ANN Training

The network diagram of the build model is found in the following figure.

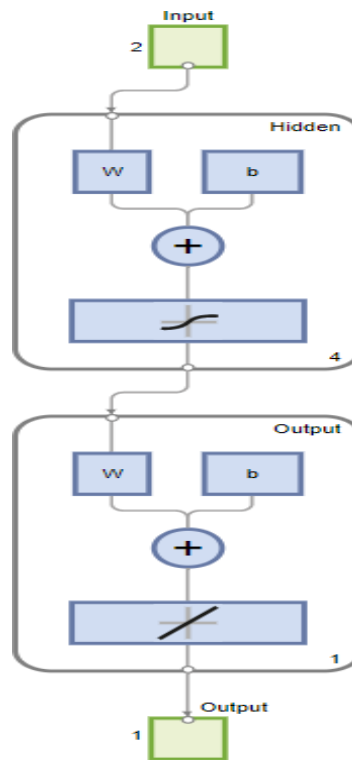


Figure 3.4: My ANN model

The training process of an artificial neural network (ANN) involves generating predictions and then refining them through the back-propagation mechanism. This iterative process occurs over a specified number of iterations as part of the training mechanism. Our dataset encompasses a total of 3000 examples, including both primary signals and noise signals. To ensure effective training, this dataset is divided into distinct subsets: 60% for training, 20% for validation, and 20% for testing. Furthermore, the dataset is fed into the ANN architecture, illustrated in Figure 3.4. Each training batch consists of 32 examples, and the network's weight adjustments are performed over numerous iterations to enhance its learning capabilities.

In the model, we have employed the back-propagation method as a pivotal technique to enhance weight optimization, thereby achieving superior performance. This strategy harnesses gradient descent, supplemented by a carefully selected learning rate of $\alpha = 0.01$, to iteratively refine the network's weights. By implementing this methodology, we strategically traverse the weight space, facilitating convergence towards more favorable local minima. Consequently, the overall efficacy of the model is significantly heightened.

ANN Testing

Testing is the last step where we check how well our sensing method works. We use a set of test data to figure out how well we can detect signals. We split the data set into smaller parts. From these smaller parts, we pick out features like CAV and Logdet.

These features go into the trained neural network. It predicts if the alternative situation H_1 is happening or not. By counting how many times the neural network correctly predicts H_1 , we can calculate how good our detection is for a specific signal strength (SNR). Similarly, we can find the likelihood of mistakenly thinking there's a signal when there isn't P_{fa} . We do this by using similar steps but with different data. Our artificial neural network helps decide whether the main signal is present or not.

Chapter 4

Results

4.1 Metrics

Utilizing the provided measurement setup, the suggested method for detecting available frequencies is examined using two characteristics mentioned earlier, namely CAV and Logdet. The achieved outcomes are analyzed to determine the most effective approach. During the assessment process, various parameters of the artificial neural network (ANN) design were considered. Subsequently, by employing the testing dataset, the probabilities of correctly identifying signals P_d are calculated.

The conducted evaluation encompassed a meticulous selection of training parameters to ensure the thorough assessment of the model's performance. The training duration spanned 100 epochs, affording an extensive exposure of the model to the training dataset and facilitating the gradual refinement of its internal representations. The learning rate, set at 0.1, assumed a pivotal role in the optimization process, determining the magnitude of weight updates during gradient descent. This parameter's careful calibration struck a balance between convergence speed and stability.

Introducing a regularization parameter, λ , of 0.001 was instrumental in combating the inherent risk of overfitting in complex models. The inclusion of L2 regularization within the optimization framework endowed the model with the ability to mitigate the undue influence of individual weights, thus enhancing its capacity to generalize to unseen data. This regularization term, combined with the learning rate, functioned as a harmonized duo in guiding the optimization landscape [49].

Within the training loop, the choice of optimization algorithm, Adam, demonstrated efficacy in navigating complex, high-dimensional parameter spaces. The training strategy of one epoch per iteration enabled a measured update of model parameters, avoiding drastic fluctuations that could hinder convergence. The use of mini-batches, each comprising 32 samples, balanced computational efficiency and convergence quality, effectively leveraging parallelism in modern hardware architectures [50].

The learning rate schedule was adopted as a strategic measure to fine-tune the model's learning process. Employing a piece-wise approach, the learning rate experienced controlled reductions every 30 epochs, thereby adapting the rate of learning as the model approached convergence. This dynamic learning rate regimen optimally balanced rapid convergence in early epochs with precision during later stages. [51]

To promote variation in input order and prevent any unintended sequence bias, the 'Shuffle' strategy was implemented to randomize the presentation of training data during each epoch. Furthermore, the 'training-progress' plot, which was facilitated by the 'Plots' option, provided a comprehensive visualization of the training trajectory, revealing valuable insights into the model's learning dynamics.

Ultimately, the collective integration of these metrics and parameters substantiated a robust evaluation framework, ensuring the model's effective convergence, generalization, and performance assessment.

4.2 Results

Utilizing the provided measurement setup, the suggested method for detecting available frequencies is examined using two characteristics mentioned earlier, namely CAV and Logdet. The achieved outcomes are analyzed to determine the most effective approach. During the assessment process, various parameters of the artificial neural network (ANN) design were considered. Subsequently, by employing the testing dataset, the probabilities of correctly identifying signals P_d are calculated.

4.2.1 Validation

During the validation phase, the accuracy of the model was assessed using the validation dataset. The neural network's predictions were generated for the validation inputs stored. These predictions were then transformed into binary classifications using a threshold of 0.5. The accuracy of these binary predictions was calculated by comparing them against the actual target labels of the validation data subset. The validation accuracy was determined as the ratio of correctly predicted instances to the total number of instances in the validation subset. The corresponding results, including the epoch number and the achieved validation accuracy, were displayed to provide insight into the model's performance during training which shows that it reached the maximum on the first epoch as you can see in the following figure.

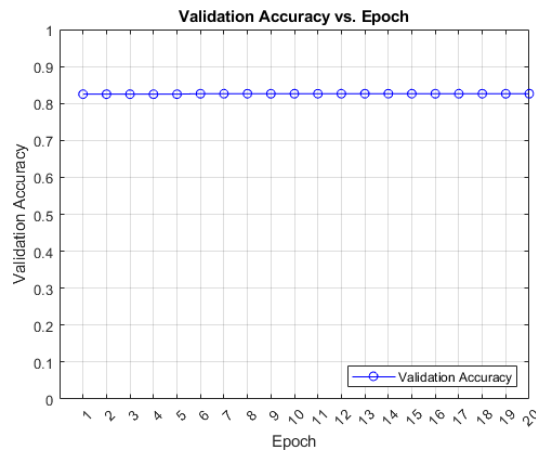


Figure 4.1: Validation Accuracy

4.2.2 ANN Training Analysis

Manipulation in Training Data

The devised mechanism employing Artificial Neural Networks (ANN) encompasses a specific architecture that necessitates effective training for accurate hypothesis classification. In alignment with the objectives of this study, a crucial imperative is to ensure the simplicity of the architecture. This is imperative to enable the practical implementation of the scheme within real-world sensing frameworks. In accordance with these considerations, deliberate data manipulation techniques were applied to the training dataset. This manipulation process aligns with the overarching goal of enhancing the model's ability to generalize while accommodating the pragmatic limitations of real-world deployment. The manipulation scheme modifies the model's parameters, which we will talk about in the subsequent discussion.

The manipulation was stated to see how the model deals with different values of signal over noise ratio (SNR) and how much does it effect on the probability of detection P_d with a probability of false alarm $P_{fa} = 0.1$. First we applied the model to SNR = -11 db that gave us the following probability of detection $P_d = 88.6\%$. This data point is represented on the Receiver Operating Characteristic (ROC) curve depicted below.

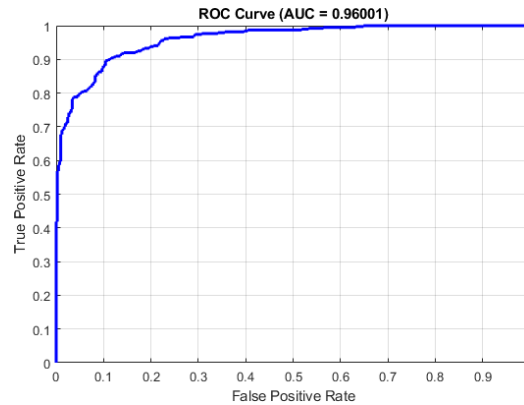
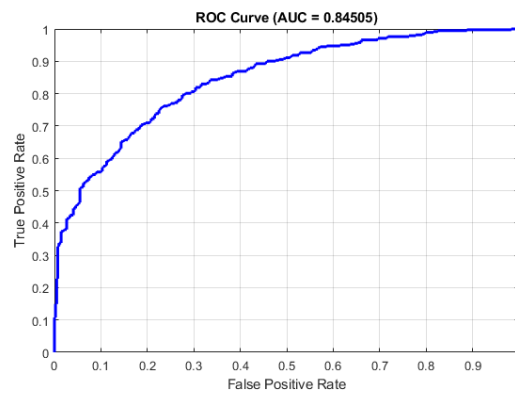
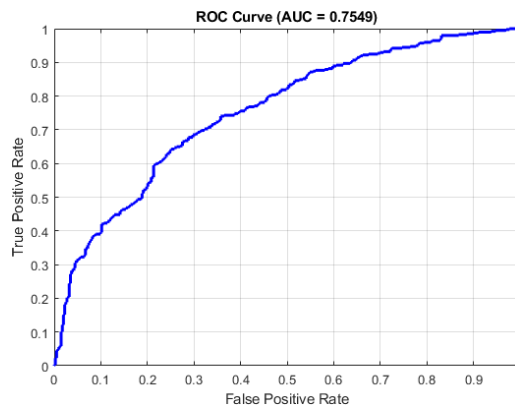


Figure 4.2: ROC of SNR= -11 db

Then we applied the model to SNR = -13 db that gave us also a good probability of detection $P_d = 80.9\%$. This forms the following Receiver Operating Characteristic (ROC) curve.

Figure 4.3: ROC of SNR= -13 *db*

After that the SNR was assigned to SNR = -15 *db* that gave us a low probability of detection $P_d = 63.3\%$. This forms the following Receiver Operating Characteristic (ROC) curve.

Figure 4.4: ROC of SNR= -15 *db*

The results will be discussed clearly in the following chapter. Another important parameter that will be subject to manipulation is the number of antennas under consideration. Currently, the signal-to-noise ratio (SNR) has been adjusted to -15 *dB*, and the investigation has shifted towards altering the number of antennas. The first exploration was illustrated by the graph in Figure 4.4, where the SNR is held at -15 *dB* and the number of antennas, denoted as M , is set to 2.

We increased the number of antennas to be $M = 4$, and we got a probability of detection $P_d = 73\%$ that is represented in the following Receiver Operating Characteristic (ROC) curve.

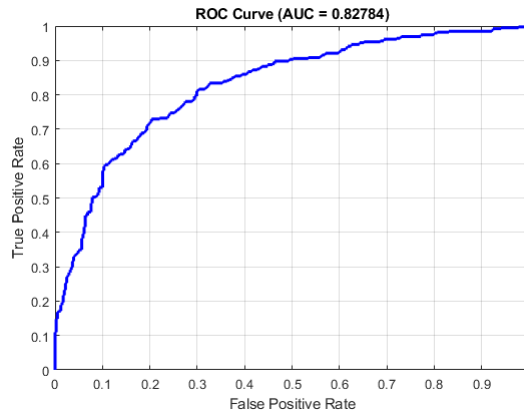


Figure 4.5: ROC with 4 antennas

The software defined radio (SDR) that is used N310 has a special advantage that it can hold up to 8 antennas. So now the following experiment will be built on $M = 8$ antennas. This showed the following results a probability of detection $P_d = 84\%$ we represented them in the following Receiver Operating Characteristic (ROC) curve.

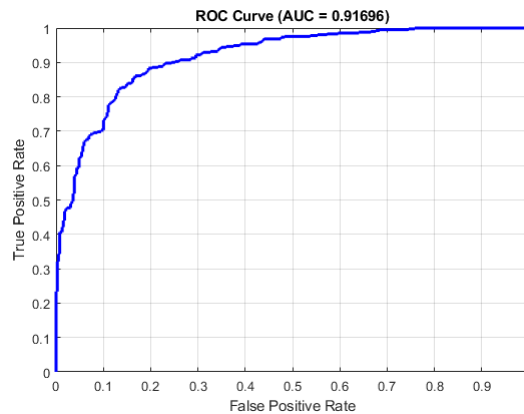


Figure 4.6: ROC with 8 antennas

4.3 Comparison with state-of-the-art

The significant outcomes we've obtained can be attributed to the use of machine learning algorithms in spectrum sensing, as emphasized in the important work of [27]. Specifically, the role of Artificial Neural Networks (ANN) stands out for delivering a notable probability of detection $P_d = 88.6\%$ in our model, surpassing other methods like energy detection, as discussed in [6].

In the research conducted by [5], it's shown that the Feedforward Neural Network excels due to its minimal time delay compared to other techniques. It's recognized for being a quick learner, a characteristic evident in our validation results where the model quickly approached the peak as shown in the Fig (4.1).

As explained in [25], using the absolute value of covariance emerges as an essential approach, especially when dealing with weak signal-to-noise ratios (SNR). This approach complements the strengths of ANN, as highlighted in [42], leading to improved results.

Furthermore, it's noteworthy that despite challenging conditions with low SNR, our model demonstrated impressive error control. This achievement can be attributed to the strategic use of the back-propagation technique, which fine-tunes the model's parameters and reduces errors. The outcome is a consistently low test error, without exceeding 0.3.

In essence, our model's robust performance is the result of a well-planned integration of ANN, the quick learning capabilities seen in Feedforward Networks, the synergy with the absolute value of covariance in low SNR scenarios, and the effectiveness of error reduction through backpropagation. This alignment with established research enriches the field of spectrum sensing methodologies.

Chapter 5

Discussion and Conclusion

5.1 Discussion

In this section we evaluate the ANN algorithm of the spectrum sensing and analyze the results. So after manipulating the SNR and testing the model on the following values [- 11, -13, -15] and having the number of antennas $M = 2$. After testing these scenarios, we got the following graph that shows the probability of detection in these different SNRs over 100 epochs.

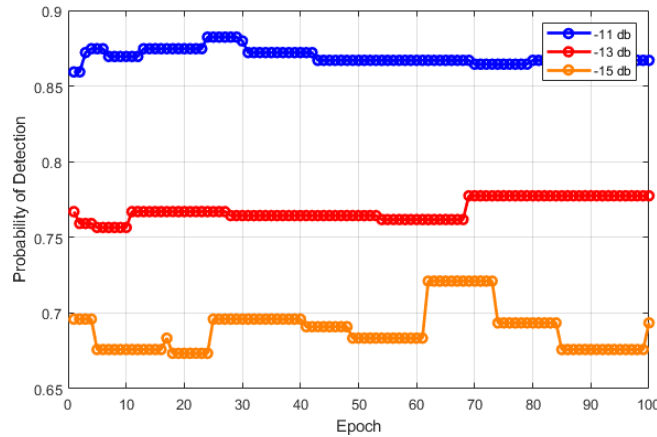


Figure 5.1: Probability of detection for different SNR

This graph showed the effect of the SNR on the probability of detection as we discussed before that when the SNR value decrease it will make it harder to detect the signal efficiently.

We also can see in the following figure the ROC curves of these different values of SNR which shows that when the Signal-to-Noise Ratio (SNR) decreases it blur the distinction between actual signal and background noise, causing the classifier to struggle in distinguishing positives from negatives. Consequently, the ROC curves shift closer to the diagonal, indicating reduced discriminative ability.

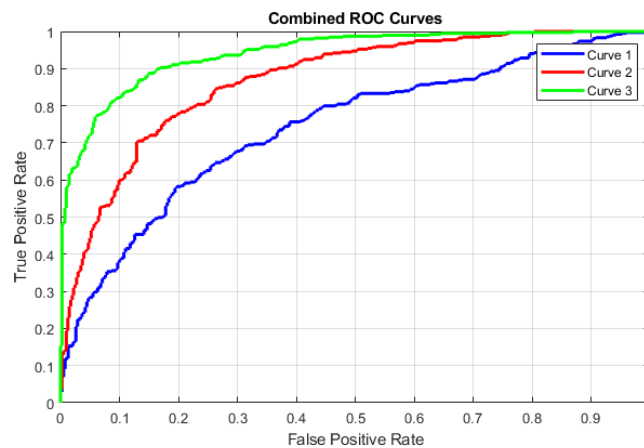


Figure 5.2: ROC for different SNR values

Then after adjusting the SNR to -15 dB and we changed the number of the antennas as following (2, 4, 8) we got the following graph that shows the probability of detection with these different number of antennas over 100 epochs.

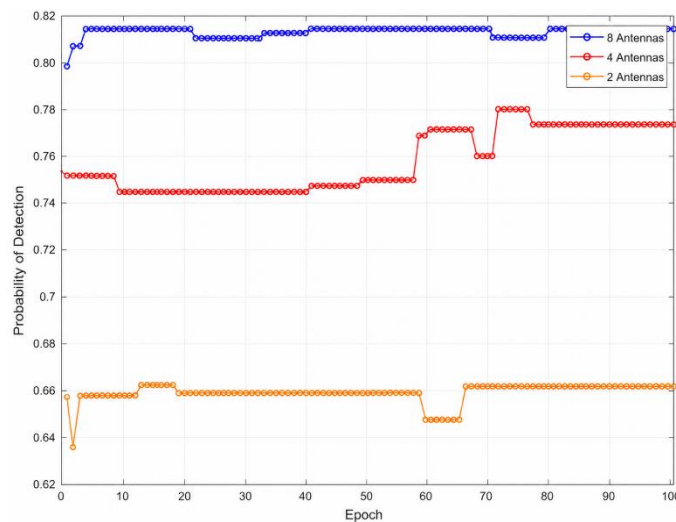


Figure 5.3: Probability of Detection for different number of Antennas

Additionally, through our experimentation with varying numbers of antennas, we noticed a significant impact on the likelihood of detection. As we increased the number of antennas, we consistently observed an associated rise in the probability of successful detection. This direct relationship emphasizes how the choice of antenna configuration can strongly influence the effectiveness of our detection methods. The more antennas we integrate, the better our system becomes at identifying and capturing signals, leading to an overall improvement in detection performance. This discovery not only highlights the practical importance of selecting the right antennas but also opens up new possibilities for enhancing our detection strategies. Looking ahead, a deeper exploration of the link between antenna quantity and detection probability will provide valuable insights for optimizing the design of more efficient and accurate

detection systems. Such insights will drive advancements in the broader fields of signal processing and wireless communication technologies.

The ROC curves presented in the following diagram with curve 1 of “2 antennas” and the rest respectively, illustrate how varying the number of antennas impacts signal differentiation from background noise. As the number of antennas grows, the distinction between the true signal and noise becomes more apparent. This enhancement leads to the classifier better recognizing the difference between positive and negative cases. As a result, the ROC curves move further from the diagonal line, signifying an improvement in the classifier’s ability to differentiate effectively.

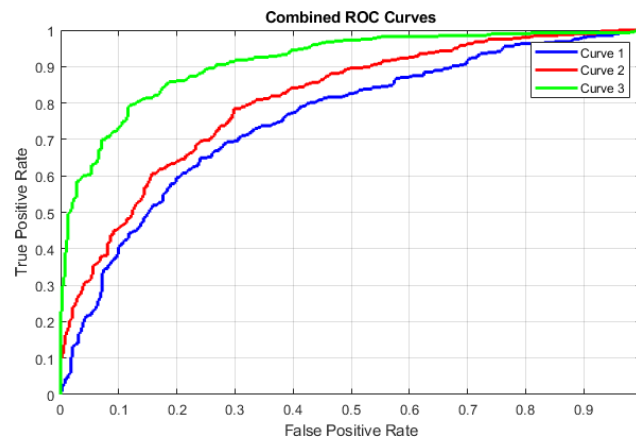


Figure 5.4: ROC for different number of antennas

5.2 Conclusion

This internship is divided into two objectives. The first objective was to improve the test bed experimentally for the software-defined radio (SDR) to enhance the signal-to-noise ratio (SNR). The second objective was to build an ANN model for better prediction of the primary user, addressing the spectrum sensing problem, which is a key component of cognitive radio. In this work, we propose an implementation of a reconfigurable SDR-based multireceiver test platform for the experimental evaluation of the ANN covariance-based model. We also discussed the signal-to-noise ratio problem. Additionally, we have detailed the different modules of the proposed platform and described the testing methodology. Furthermore, we implemented the ANN algorithm in the proposed platform to assess its performance with different SNRs and a different number of antennas. The comparison showed that in scenarios where the noise is high and the SNR is low, increasing the number of antenna receivers and using artificial intelligence methods helps increase the probability of detection.

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